

AHURUEZENMA DAVIDSON CHIEMEZUO

AI ENGINEER | AGENTIC AI DEVELOPER | AUTOMATION SPECIALIST

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PROFESSIONAL PROFILE

AI Engineer with 2+ years of hands-on experience designing and delivering production AI systems, retrieval-augmented generation pipelines, and multi-channel automation. I build systems that connect models to real operational workflows: retrieval, validation, persistence, notifications, observability, and safe human review. My work combines Python, LangChain, ChromaDB, Gemini, n8n, Supabase, AWS, and Claude Code with reliability patterns such as retries, circuit breakers, timeouts, and dual-channel alerting.

Target roles: AI Engineer, AI Developer, Agentic AI Engineer, RAG Engineer, and AI Automation Engineer. Open to permanent remote roles, AI startups, and teams working with US-compatible hours.

TECHNICAL CAPABILITIES

- AI systems: RAG architecture, semantic retrieval, vector embeddings, multi-agent orchestration, prompt engineering, multi-LLM integration, model evaluation, MCP, LangChain/LangGraph, CrewAI/AutoGen.
- Engineering: Python, TypeScript, React, FastAPI, SQL, PostgreSQL/MySQL, SQLAlchemy, REST APIs, webhooks, structured validation, test-driven development, GitHub Actions CI.
- Platforms and automation: n8n, Supabase, AWS S3, AWS CLI, CloudWatch, Gemini, Groq, OpenRouter, ChromaDB, Hugging Face embeddings, Playwright, Chrome extensions.
- Reliability and governance: retries with exponential backoff, circuit breakers, timeouts, request tracing, health checks, human-in-the-loop gates, GDPR, NDPR, ISO 27001, privacy impact assessments.

PROFESSIONAL EXPERIENCE

Freelance AI Automation Specialist

Independent Contractor — Zenko Limited, Theldoforce, DGI Integrated Services

2024 – Present | Remote

- Own delivery of three production automation systems spanning DTC lead generation, security-services operations, and AWS infrastructure monitoring.
- Built the Zenko lead pipeline from website webhook and Excel audit backup through Python validation, Gemini personalization, Supabase persistence, and automated email delivery.
- Built Theldoforce's operations system to distribute schedules, security alerts, and task assignments across Telegram, Gmail, and WhatsApp with AI-powered role tiering.
- Built DGI's AWS monitoring workflow with secured webhook capture, AI-assisted error classification, immediate Telegram alerts, and circuit-breaking for repeated failures.
- Standardized a reliability layer across client systems: 3x retries, exponential backoff, AI processing timeouts, exact failed-step alerts, and consecutive-failure pausing.

Stack: n8n · Python · Gemini · Supabase · AWS · Telegram · Claude Code

AI Engineer

ViZO Technology Limited

April 2024 – Present | Hybrid, Abuja

- Architected a production RAG system that uses document chunking, vector embeddings, semantic search, retrieval controls, and grounded model responses over organizational knowledge.
- Integrated RAG patterns with Supabase-backed storage and n8n orchestration to make AI workflows repeatable and operationally maintainable.
- Designed AI automation workflows and translated technical opportunities into adoption and ROI recommendations for leadership.
- Documented AI workflows and governance controls aligned with NDPR, GDPR, and ISO 27001; trained teams on responsible AI use and prompt engineering.

Stack: Python · LangChain · RAG · Supabase · n8n · Vector Embeddings

AWS Cloud & Automation Intern

DGI Integrated Services

May 2023 – April 2024 | Remote

- Automated cloud infrastructure provisioning and monitoring using the AWS Management Console and CLI.
- Designed structured data-ingestion workflows supporting tooling, dataset versioning, and repeatable operations.
- Deployed scalable AWS S3 storage architectures with an emphasis on fault tolerance and operational clarity.

Stack: AWS S3 · AWS CLI · CloudWatch · n8n

FEATURED AI ENGINEERING PROJECTS

Selected work ordered for AI Engineer, RAG, agent, and automation roles. Links are clickable in the PDF.

1. AI Job Application Agent

Production-hardened agentic workspace for researching job postings, analyzing candidate fit, tailoring resumes and cover letters, and preparing browser actions through a FastAPI backend and authenticated Chrome extension.

- Orchestrates agent workflows with LangGraph and supports Gemini, Groq, and OpenRouter fallback paths.
- Added Google OAuth claim validation for audience, subject, email, and email_verified before issuing application JWTs.
- Implemented user-scoped SQLAlchemy persistence for profiles, applications, documents, and job records with SQLite/Postgres readiness.
- Added request IDs, structured errors, liveness/readiness checks, upload limits, retention cleanup, and strict extension/API contracts.
- Maintains a manual-review safety gate: dry runs never submit and prepared applications remain READY_TO_SUBMIT until a human verifies the final action.
- Verified with 111 backend tests plus 7 extension build/configuration tests, Ruff, and strict mypy.

Stack: FastAPI · LangGraph · SQLAlchemy · Playwright · Google OAuth · Chrome Extension · Gemini · Groq

GitHub: github.com/DavidGasol/Job-Agent

2. VIZO-RAG System

Source-grounded retrieval assistant for organizational knowledge. The system ingests Markdown content, creates embeddings, persists vectors, retrieves evidence, and generates answers that refuse to guess when evidence is missing.

- Uses ChromaDB with all-MiniLM-L6-v2/Hugging Face embeddings and configurable retrieval controls.
- Supports query expansion for broad questions, conversation memory, and evidence-aware response generation with Gemini.
- Includes a 20-test offline suite that runs without an API key, making retrieval behavior repeatable during development.
- Documents ingestion, retrieval, prompting, evaluation, health checks, and operational usage for handoff.

Stack: Python · LangChain · ChromaDB · Gemini · Hugging Face Embeddings

GitHub: github.com/DavidGasol/VIZO-RAG

3. HealthRAG — Nigeria-First Health Triage

Synthetic-only health triage prototype combining deterministic safety rules with retrieval-grounded educational answers. It is explicitly fail-closed: it refuses real patient input and does not diagnose or prescribe.

- Checks 11 emergency and urgent-signal rules before retrieval runs, ensuring safety signals take priority over generation.
- Uses evidence citation gates and Pidgin-friendly phrasing for accessible, grounded educational responses.
- Ships with an offline release gate covering 55 tests and 16 synthetic evaluation cases without an API key.
- Separates safety, privacy, retrieval, evaluation, documentation, and operational concerns into a maintainable repository.

Stack: Python · YAML · RAG · Safety Rules · Synthetic Evaluation

GitHub: github.com/DavidGasol/HealthRAG

4. AI Restaurant Recommendation Engine

Hybrid AI recommendation system blending collaborative and content-based filtering with location and time awareness — a FastAPI backend with a React frontend, split out of the NutriDine product line.

- Combines collaborative + content-based filtering in scikit-learn for personalized, location- and time-aware restaurant suggestions.
- FastAPI endpoints for recommendations, cuisine suggestions, user profiles, and interaction tracking.
- Verified with 33 backend tests plus a frontend build in CI.

Stack: FastAPI · React · scikit-learn · SQLAlchemy · CI

GitHub: github.com/DavidGasol/restaurant-recommender

5. AI Nutrition Advisor

RAG nutrition advisor providing evidence-based, personalized guidance for Nigerian and international foods — allergen detection, ChromaDB semantic search, and Groq/Gemini backends with a rule-based offline path.

- Retrieves from a Nigerian food knowledge base via ChromaDB vector search with semantic matching.
- Detects allergens with severity levels and hidden sources; tailors advice to health goals and dietary preferences.
- Multi-LLM support (Groq + Gemini) with a rule-based analysis path that works fully offline without API keys.
- CI-enforced offline-loading guarantee plus a rule-based health-analysis test suite.

Stack: Python · ChromaDB · Groq · Gemini · Rule-based fallback

GitHub: github.com/DavidGasol/ai-nutrition-advisor

6. NutriVoice — Multilingual Voice Nutrition Agent

Voice-first AI agent for nutrition consultations, appointment booking, and healthy food ordering in Nigeria — speaking English, Igbo, Hausa, and Yoruba.

- Live, speech-driven consultations with real-time transcripts and audio visualization.
- Multilingual support across English, Igbo, Hausa, and Yoruba for broader accessibility.
- Connected appointment booking and healthy food ordering flows with location awareness.

Stack: React · TypeScript · Vite · Gemini
GitHub: github.com/DavidGasol/nutrivoice

7. Financial Statement Extraction Pipeline

Document-processing pipeline for scanned Nigerian financial statements. It converts OCR input into structured, validated data and supports confidence scoring, detection, parsing, and export workflows.

- Covers OCR ingestion, number parsing, field detection, validation, confidence scoring, and exporters.
- Uses focused modules and tests for parsing, detection, validation, logging, OCR, jobs, and export behavior.
- Provides a repeatable foundation for financial-document processing rather than a one-off extraction script.

Stack: Python · OCR · Pydantic · pytest · GitHub Actions
GitHub: github.com/DavidGasol/financial-statement-pipeline

8. R-Medy — Natural Health Solutions

Production web application for a natural-health brand with a modern Next.js/TypeScript frontend, AI-assisted lead capture, secure webhook integration, deployment documentation, and a public production site.

- Built responsive product and content experiences with TypeScript, Tailwind, and Next.js conventions.
- Connected lead capture to secure webhook-based workflows and documented deployment/security considerations.
- Public site: rmedy.qzz.io. Client-facing delivery demonstrates the ability to ship beyond prototypes.

Stack: Next.js · TypeScript · Tailwind CSS · Webhooks · Deployment
GitHub: github.com/DavidGasol/R-Medy
Live: rmedy.qzz.io

ADDITIONAL PRODUCTION & AUTOMATION WORK

Supporting projects that demonstrate client delivery, infrastructure automation, data engineering, and operational reliability.

9. Lead Generation & Personalization Automation — Zenko Limited

End-to-end DTC lead pipeline from website webhook to auditable backup, Python validation, Gemini personalization, Supabase persistence, and automated email delivery.

- Uses an Excel/Google Drive backup before validation to preserve an operational audit trail.
- Adds 3x automatic retry with exponential backoff, a 30-second AI timeout, Telegram alerting, and circuit-breaker logic.
- Designed as a production workflow with failure visibility instead of a simple happy-path integration.

Stack: n8n · Python · Gemini · Supabase · Google Drive · Email API

10. Operations & Personnel Alert System — Theldoforce

Bi-directional operations automation for security services, distributing schedules, security alerts, and task assignments across Telegram, Gmail, and WhatsApp.

- Uses AI-powered tiering by recipient role so each team member receives relevant detail instead of broadcast noise.
- Includes dual-channel failure alerting with exact failed-step context for faster diagnosis.
- Automates real-time personnel scheduling and alert management across three communication channels.

Stack: n8n · Python · AI Model · Supabase · Telegram · Gmail · WhatsApp

Workflow repository: github.com/DavidGasol/theldoforce

11. AWS Infrastructure Error Monitoring — DGI Integrated Services

Fault-tolerant monitoring workflow that receives secured AWS error signals, classifies them with AI assistance, and pushes immediate Telegram alerts to operations teams.

- Moves incident detection from manual log review toward near-real-time notification.
- Combines AI classification with retry, timeout, and consecutive-failure circuit-breaking patterns.
- Designed for operational clarity: the alert includes the failed step and actionable context.

Stack: AWS API · n8n · Python · AI Model · Supabase · Telegram API

12. AI Email & Task Assistant

Personal AI assistant in n8n that triages email, manages tasks, and sends daily briefings via Telegram — powered by Google Gemini with a Google Tasks/Calendar/Drive tool suite.

- Watches Gmail with sender/keyword filtering and evaluates follow-ups before acting.
- Runs a Gemini agent with a tool suite (Tasks, Gmail, Drive, Calendar) and Supabase-backed conversation memory.
- Automates a daily TODO, transaction alerts, and a Monday weekly report to Telegram + Google Sheets.

Stack: n8n · Gemini · Telegram · Google Tasks · Supabase

GitHub: github.com/DavidGasol/ai-email-task-assistant

13. AI Academic Assignment Generator

n8n workflow that turns a short Telegram query into a fully formatted academic assignment — Gemini-generated content converted to a strict HTML document and saved to Google Drive/Sheets.

- Telegram intake to Gemini agent to HTML formatting (12pt Times New Roman, double spacing, 1-inch margins).
- Automatic record-keeping: every assignment is uploaded to Google Drive and logged to Google Sheets.

Stack: n8n · Gemini · Telegram · Google Drive · Google Sheets

GitHub: github.com/DavidGasol/ai-academic-assignment-generator

14. AI-News-Update — Automated News Digest

A fully automated daily AI news digest. This is intentionally last: it demonstrates scheduled automation, data freshness, deduplication, delivery, deployment, and CI reliability after the core AI engineering work above.

- Runs from a GitHub Actions cron at 6 AM, fetches RSS sources, and publishes a public dashboard on Vercel.
- Seeds cross-day URL/title deduplication from a committed archive and retains 180 days of history.
- Sends multi-recipient email digests and includes CI tests for deduplication, archive merging, and retention behavior.
- Live dashboard: ainl.vercel.app.

Stack: Python · GitHub Actions · feedparser · SMTP · Vercel

GitHub: github.com/DavidGasol/AI-News-Update

Live: ainl.vercel.app

CERTIFICATIONS

- Claude Code 101 — Anthropic Education (2026)
- Introduction to Subagents — Anthropic Education (2026)
- Introduction to Agent Skills — Anthropic Education (2026)
- AI Fluency: Framework and Foundations — Anthropic Education (2026)
- Claude 101 — Anthropic Education (2026)
- Workflow Automation Training — Udemy (2025)
- Data Protection Officer Training — Data Protection Academy (2024)
- AWS S3 Fundamentals — AWS Training (2022)

EDUCATION

Bachelor of Science, Sociology and Anthropology — Obafemi Awolowo University, Ile-Ife (2023).

Academic foundation in systems theory, organizational behavior, and qualitative research applied to human-AI interaction and agent workflow architecture.